

Introduction to Data Mining

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Course Introduction

- Course Coordinator:

Ting Kai Ming

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Outside class time, to meet the course coordinator face-to-face to discuss any issues related to this course, please send an email to make an appointment.

Course Objectives

On the completion of this course, students should be able to

- Describe the motivation and the need of data mining.
- Demonstrate an understanding of different data mining tasks and the basic principles of different data mining techniques.
- Understand various complex data objects associated with real world problems and their data mining tasks and techniques.
- Apply problem solving skills in using appropriate data mining techniques to a target problem.
- Research and analyze the advantages and disadvantages of competing data techniques.
- Demonstrate an understanding of current major research issues in the field.

Course Introduction

- Course Organisation:

Workshop: 2 hours per week

References:

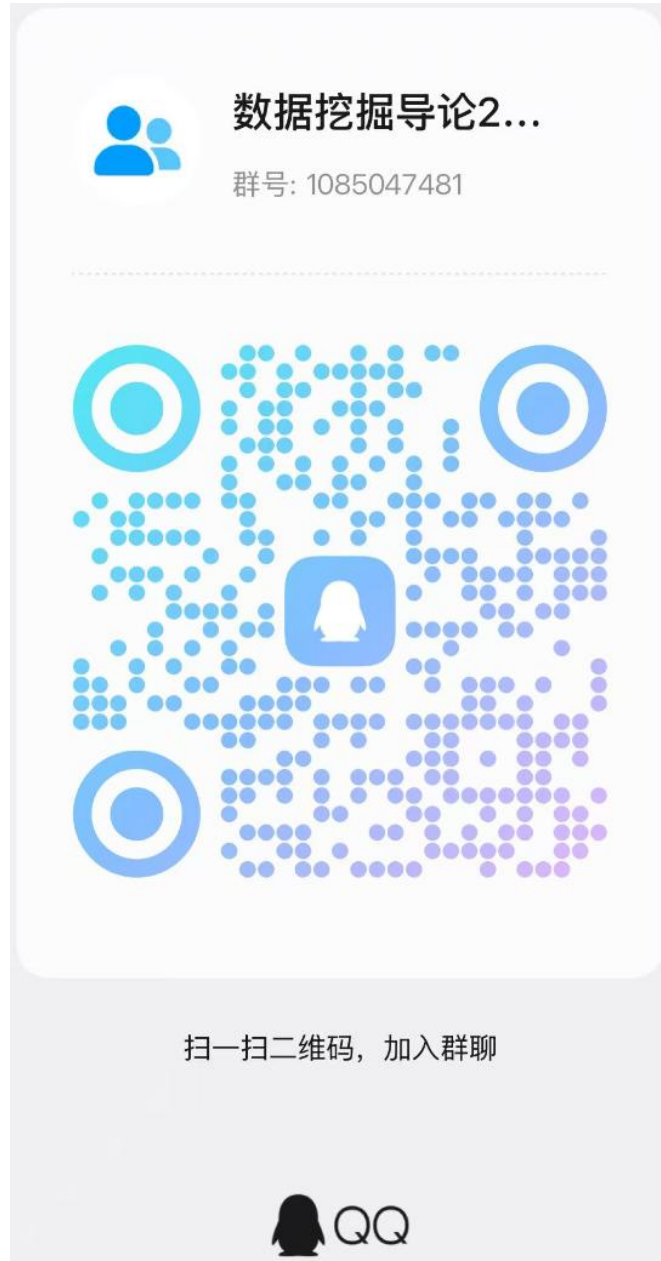
- C. C. Aggarwal, Data Mining: The Textbook. Springer, 2016.
- P.-N. Tan, M. Steinbach, A. Karpatnu and V. Kumar, Introduction to Data Mining, Pearson, 2nd edition, 2018.
- I. H. Witten, E. Frank, Mark. A. Hall, Christopher J. Pal. Data Mining: Practical Machine Learning Tools and Techniques, Morgan Kaufmann, 3rd edition 2016.

Assessment

Three components

1. In-class assessment (20%)
2. One assignment (40%)
3. A final examination (40%)

Announcements & Online Delivery in QQ



Preparation for workshop/In-class assessment

- During the workshop, students are to engage in the discussion and answer the questions listed on the slides or raise additional questions for class discussion.
- This means that students must have read the study materials for the week before attending the workshop.
- In-class assessment: Presentation for your preparation to complete the group assignment.

Study Plan

Week#1-4	Introduction, Data, Distances
Week#5	Data Mining Algorithms
Week#6-11	Time series, Trajectory, Graph
Week#12-16	Data Streams, Social Networks, Web Data
Group Assignment due	6 June 2026

Advice on study for this course

- You are expected to know the key concepts (assess in the final examination)
- If you are overwhelm by many details, ask yourself whether you understand the key ideas.
 - You shall understand the key ideas first, before trying to understand the details. The details are less important than the key ideas.
- **Ask questions.**
- **Seek help** if you feel you are not progressing well in this course. Talk to the course coordinator; and your fellow students.

Reading for week 1:

- Required reading: Aggarwal Chapter 1
- Supplementary reading: Tan Chapter 1

Acknowledgement:

Unless stated otherwise, the images on the following slides are sourced from C. C. Aggarwal, Data Mining: The Textbook. Springer, 2016.

Why Data Mining?

- Data is collected for many daily activities, e.g., purchases in a supermarket or online platform, browsing in WWW, asking questions to LLM, etc
- Data is an asset for organizations.
- But raw data is not useful.
- Data shall be “mined” to turn raw data into useful information for various purposes.
- There is a potential to use AI agent to accomplish some of these data mining tasks, but understanding the underlying principles is paramount.

Data Mining: Two Definitions

“Data mining is a process of extracting previously **unknown**, **valid**, and **actionable** information from large databases and then using the information to make crucial **decisions**.”

-- Cabena, P., Hadjinian, P., Stadler, R., Verhees, J. & Zanasi, A. (1997)

“Data mining is the study of collecting, cleaning, processing, analyzing, and gaining useful insights from data. “

-- Aggarwal, C.C. (2016)

Four Data Mining Tasks/Problems

- Classification
- Clustering
- Anomaly detection
- Association/Frequent pattern mining

Classification: Problem definition

Definition 1 (Data Classification) *Given an $n \times d$ training data matrix D (database D), and a class label value in $\{1 \dots k\}$ associated with each of the n rows in D (records in D), create a training model M , which can be used to predict the class label of a d -dimensional record $Y \notin D$.*

Table of baby-name data
(baby-2010.csv)

	name	rank	gender	year
Class	Jacob	1	boy	2010
Positive	Isabella	1	girl	2010
Negative	Ethan	2	boy	2010
Negative	Sophia	2	girl	2010
Positive	Michael	3	boy	2010
Positive				

Field names

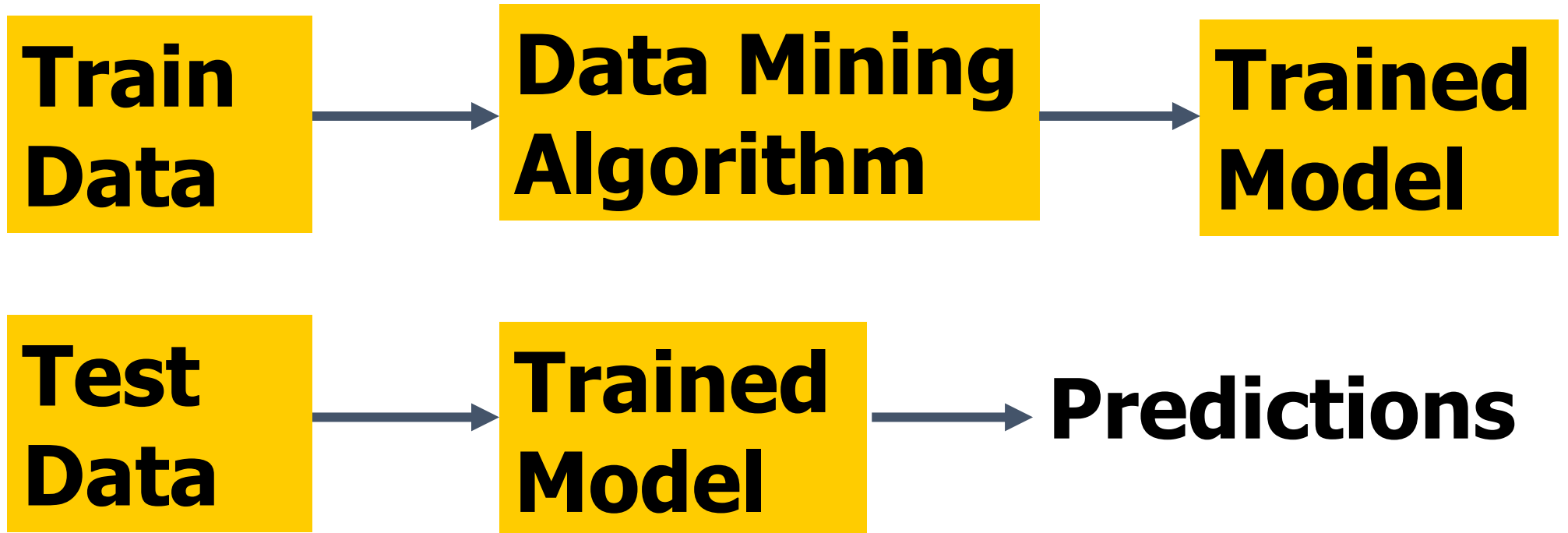
One row (4 fields)

2000 rows all told

A regression problem is similarly defined, except that the categorical class attribute is replaced with a real-valued attribute

Training and Testing processes

Classification or regression is a supervised learning task



Classification methods

- Support Vector Machines
- Neural networks
- K nearest neighbor classifier
- Decision Trees
- Rule-based classifier

Example Rules: If A then class positive

If B then class negative

Discussion #1

1. Classification and regression methods are usually described as supervised learning methods. Explain why. And describe the difference between the two.
2. What do you need in order to produce a classification model?
3. How do you know that you have produced a good model?
4. Why is a classification model also called a generalized model?

Clustering: Problem definition

Definition 2 (Data Clustering) *Given a data matrix D , partition its rows (records) into sets/clusters $C_1 \dots C_k$, such that the rows (records) in each cluster C are “similar” to one another.*

**This is an unsupervised learning task because no labels are provided in a dataset.
And no explicit training and testing processes.**

Clustering methods

- K-means/k-medoids
- DBSCAN
- Spectral clustering
- Gaussian mixture model
- Deep learning based clustering
- IDK (Isolation Distributional Kernel) based Clustering

Discussion #2

1. Explain how a clustering task differs from a classification task.
2. What is the core operation/measurement required in a clustering task?
3. How do you know that you have produced a good clustering outcome?

Outlier detection: Problem definition

Definition 3 (Outlier Detection) *Given a data matrix D , determine the ***few*** rows of the data matrix that are very ***different*** from the ***majority*** of rows in the matrix.*

Other names: anomaly detection, novelty detection

Outlier detection methods

- Distance-based anomaly detectors
- Local Outlier Factor (LOF)
- Isolation-based anomaly detectors:
 - Isolation Forest, IDK anomaly detector
- One-class Support Vector machine (OCSVM)
- Deep learning based anomaly detectors

Discussion #3

1. Explain how an anomaly detection task differs from a classification task.
2. Explain how it differs from a clustering task.
3. How do you know that you have produced a good anomaly detection outcome?

Frequent pattern mining: Problem Definition

Definition 4 (Frequent Pattern Mining) *Given a binary $n \times d$ data matrix D , determine all subsets of columns such that all the values in these columns take on the value of 1 for at least a fraction s of the rows in the matrix. The relative frequency of a pattern is referred to as its support. The fraction s is referred to as the minimum support.*

Definition 4.1 (Association Rules) *Let A and B be two sets of items. The rule $A \Rightarrow B$ is said to be valid at support level s and confidence level c , if the following two conditions are satisfied:*

- 1. The support of the item set A is at least s .*
- 2. The confidence of $A \Rightarrow B$ is at least c .*

Association/Frequent Pattern Mining methods

- Enumeration-tree algorithms
- Apriori algorithm

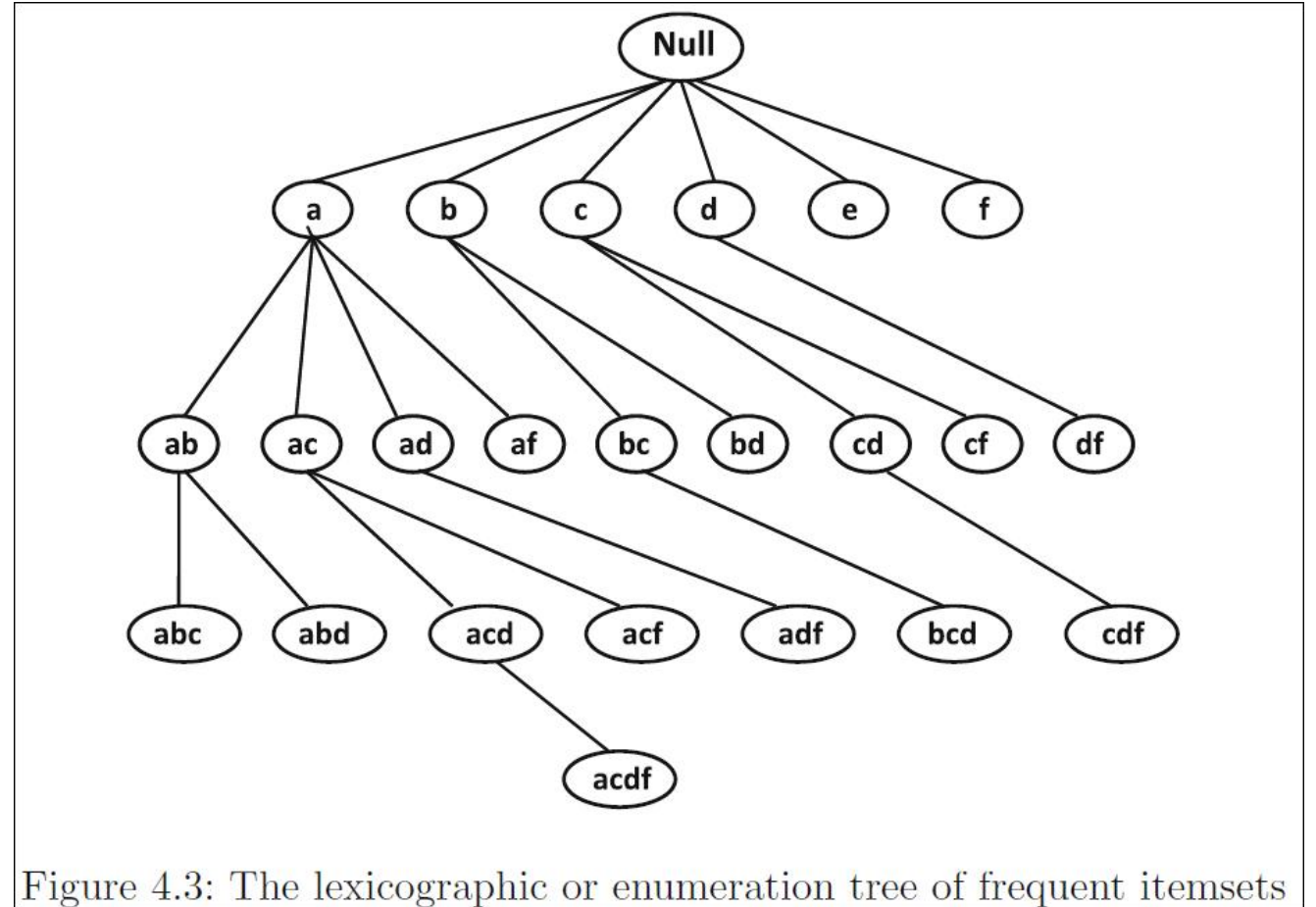


Figure 4.3: The lexicographic or enumeration tree of frequent itemsets

Discussion #4

- How does association pattern mining differ from other data mining tasks?
- How does association pattern mining differ from rule-based classification?

Multidimensional data

Often represented as multi-dimensional vectors

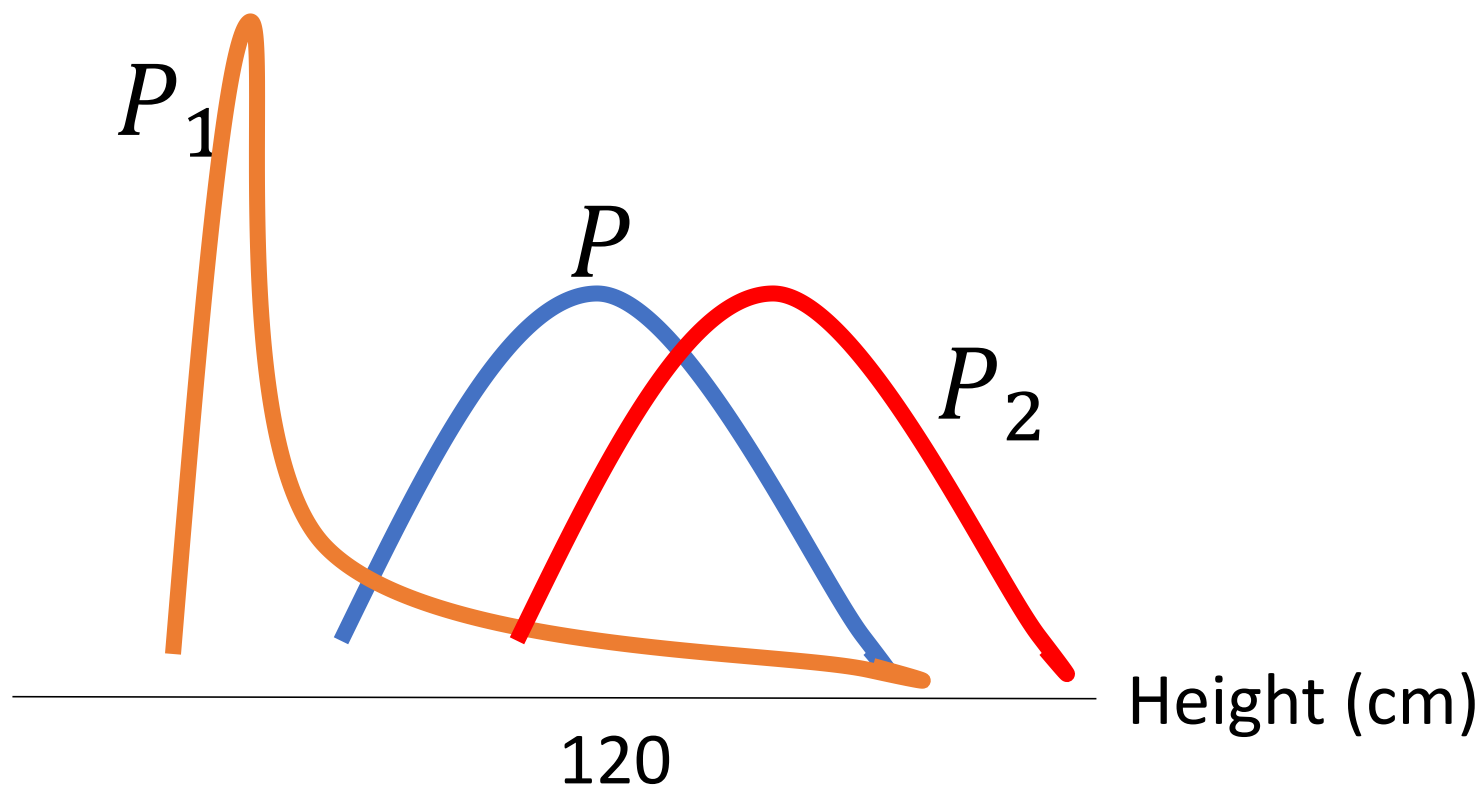
<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...

Assumption:

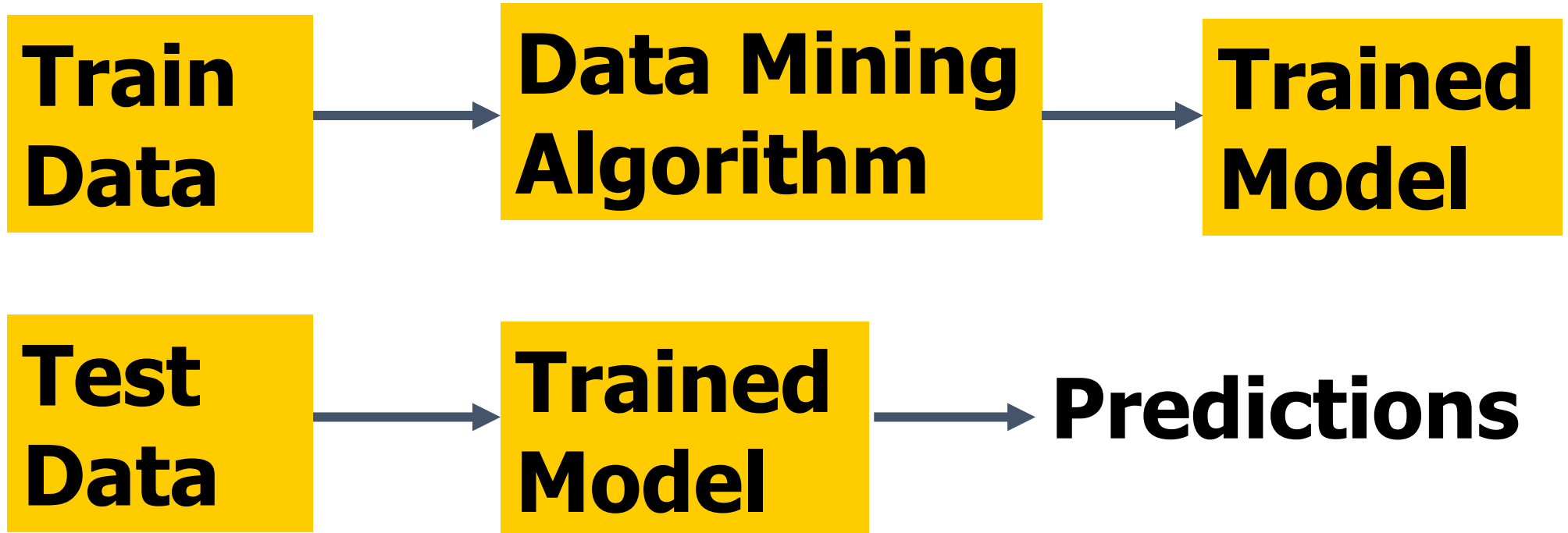
Every data point in a dataset is **independent and identically distributed (i.i.d.)**.

In other words, the points x are assumed to have been generated from the same (unknown) distribution P , i.e., $x \sim P$

What does i.i.d. mean to you?



Is there any issue in using a classification model trained from P to classify a test set derived from $P_1 \neq P$?



Multidimensional data versus Complex data objects

- Data has been conventionally prepared in a multidimensional format in a vector representation which assumes each vector (or data point) is independent and identically distributed (i.i.d.).
- Complex data objects are not i.i.d., but dependency-oriented data. Examples are: text, graph/network, sequences, spatial data objects, time series.

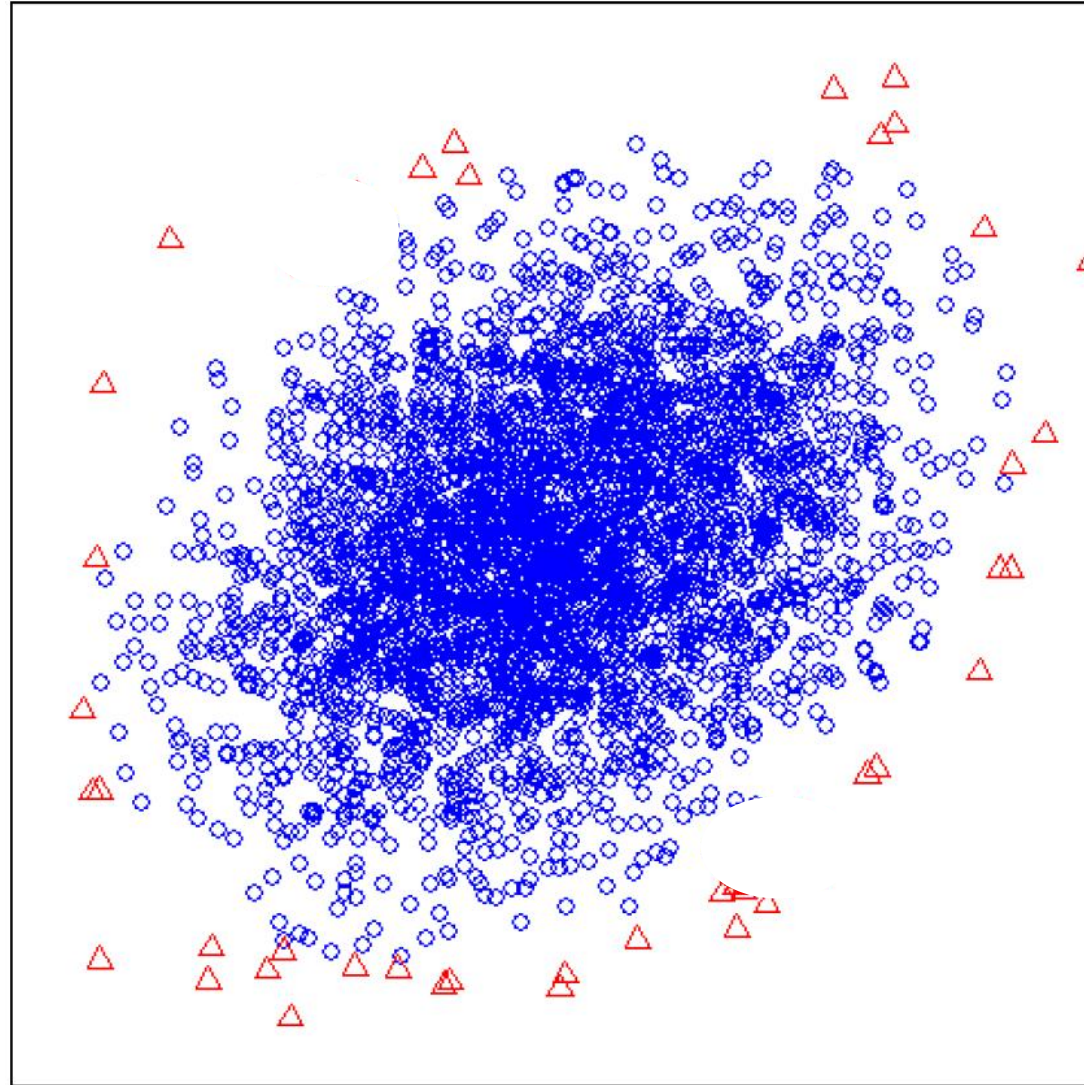
The concept of independent and identically distributed (i.i.d.) is important because

- a) A multidimensional dataset is assumed to be i.i.d.; or all points in this dataset are generated from the same (unknown) distribution.
- b) Many data mining algorithms have been developed for multidimensional data which has the i.i.d. assumption.
- c) Complex data objects are “complex” because they are not i.i.d., but dependency-oriented data.
- d) One major approach of dealing with complex data objects is to convert the data with a complex data type into multidimensional data.

Impact of Complex Data Types on Problem Definitions

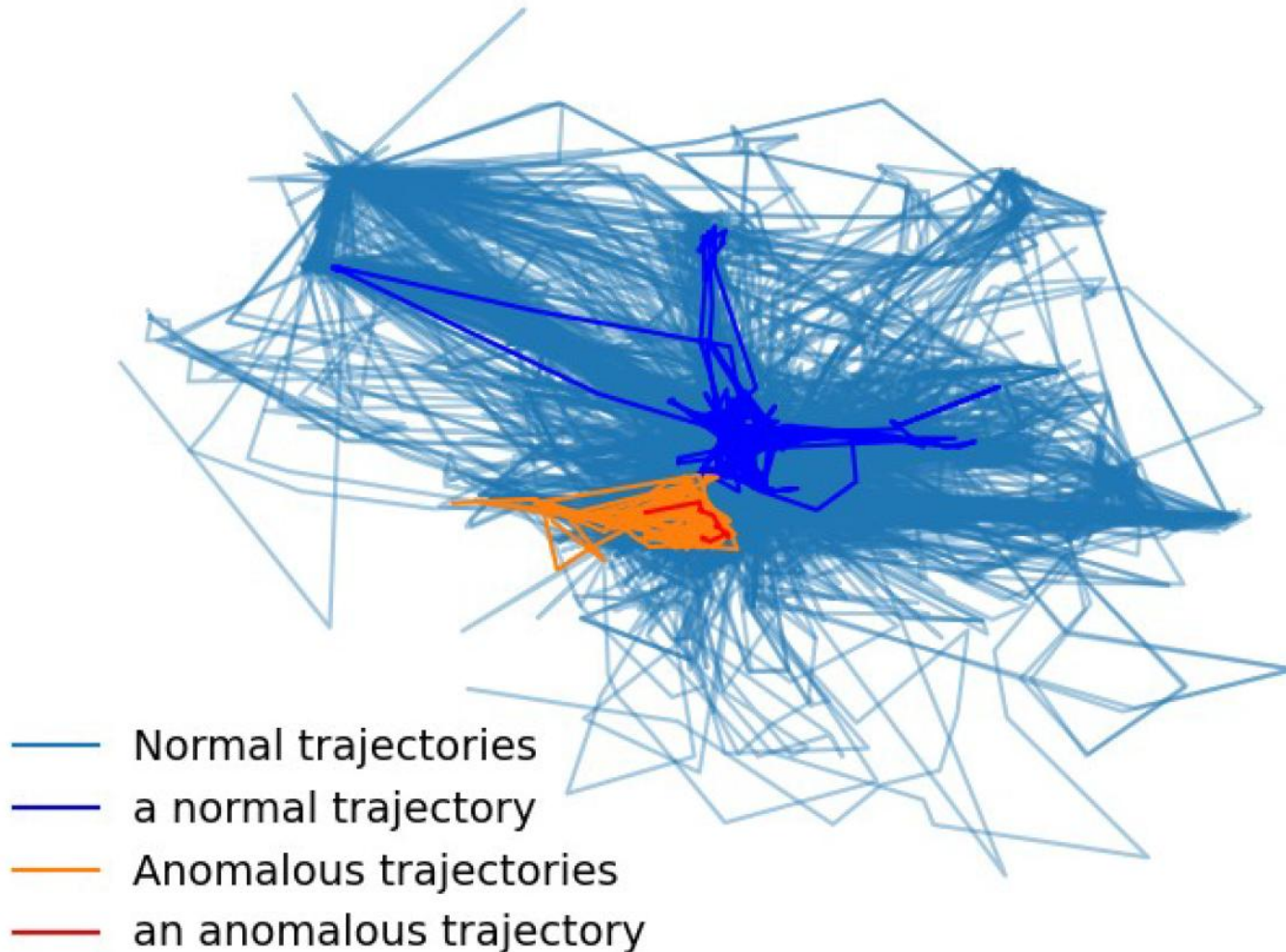
Problem	Time series	Spatial	Sequence	Networks
Patterns	Motif-mining Periodic pattern	Colocation patterns	Sequential patterns Periodic Sequence	Structural patterns
	Trajectory patterns			
Clustering	Shape clusters	Spatial clusters	Sequence clusters	Community detection
	Trajectory clusters			
Outliers	Position outlier Shape outlier	Position outlier Shape outlier	Position outlier Combination outlier	Node outlier Linkage outlier Community outliers
	Trajectory outliers			
Classification	Position classification Shape classification	Position classification Shape classification	Position classification Sequence classification	Collective classification Graph classification
	Trajectory classification			

Example 1: Point Anomalies

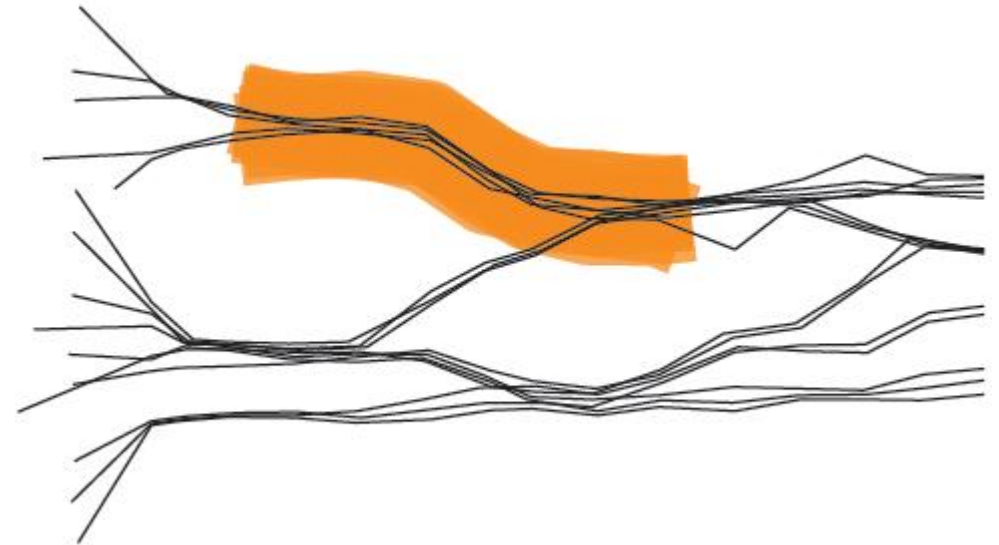


Example 2: Trajectories

(a) Tracks of Sheep-dogs & sheep

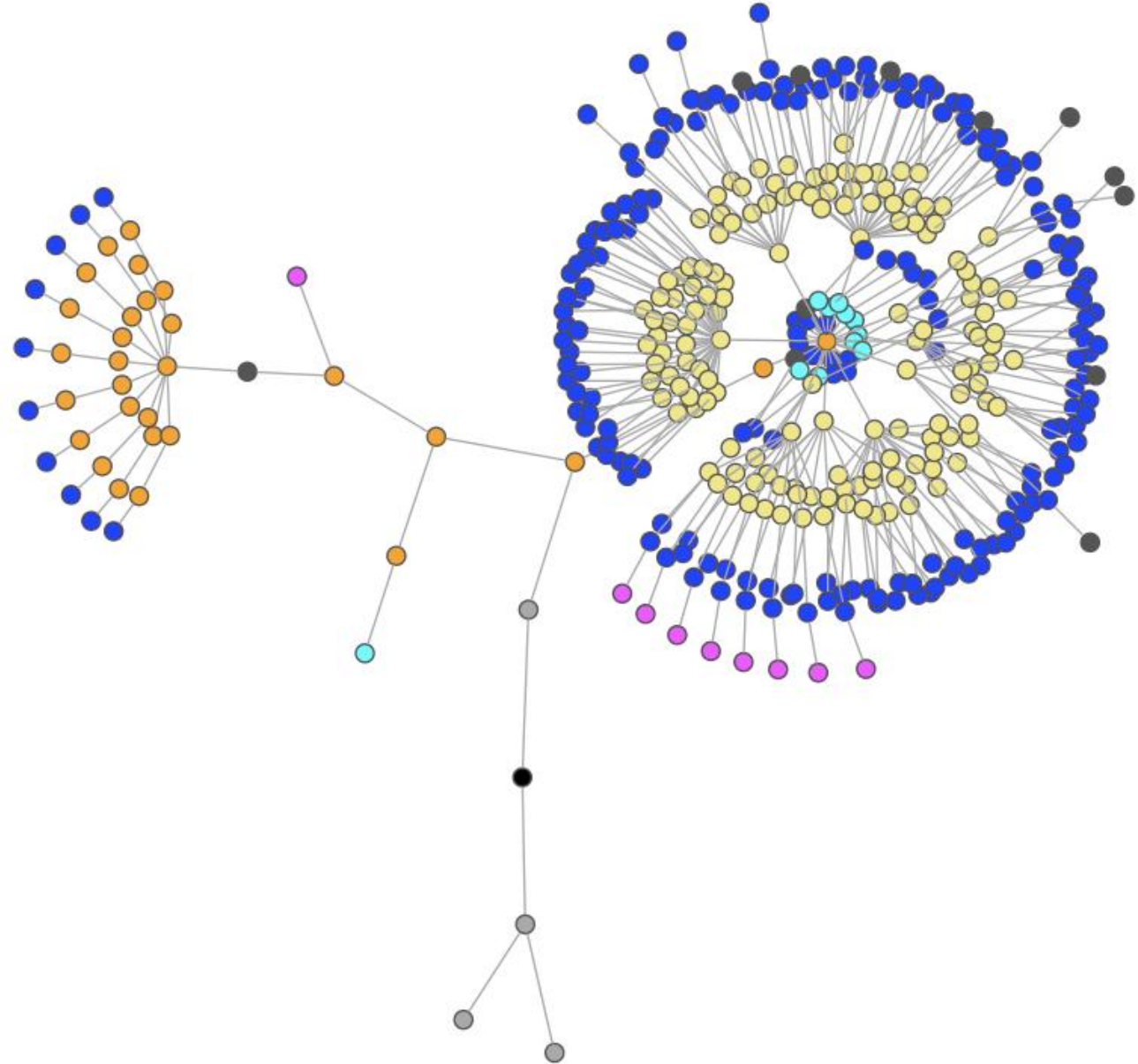


(b) Flight paths of some birds



Source: <https://www.uu.nl/en/research/algorithms/computational-geometry/research-themes/trajectory-data-analysis>

Example 3: Anomalous subgraphs in a Network



Outlier detection: Problem definition

Definition 3 (Outlier Detection) *Given a data matrix D , determine the ***few*** rows of the data matrix that are very ***different*** from the ***majority*** of rows in the matrix.*

Discussion:

- 1. Explain what an anomaly is in Example 1.*
- 2. In Example 2, explain what makes a trajectory anomalous.*
- 3. In Example 3, explain what makes a subgraph anomalous.*

Different types of learning methods

- Supervised/Inductive learning: Classification
- Unsupervised learning: Clustering, outlier detection
- Semi-supervised learning
- Transductive (a special semi-supervised) learning

Questions:

1. Does a semi-supervised learning refer to supervised or unsupervised learning or something completely different?
2. How does semi-supervised learning impact on model evaluation?

Week 1 Summary

1. Data mining is a process of extracting previously **unknown**, **valid**, and **actionable** information from large databases and then using the information to make crucial **decisions**.
2. Data mining tasks can be categorized into four main categories: Classification, Clustering, Anomaly Detection and Frequent pattern mining.
3. The "complexity" of complex data objects is with reference to (conventional) multidimensional data, assumed to be independent and identically distributed (i.i.d.). Each complex data type varies from the multidimensional data in different ways which affect its task/problem definition.
4. Many of the existing algorithms have the i.i.d. assumption and assume multidimensional data. One approach in dealing with a complex data type is to convert the data such that it complies with these assumptions, then these existing algorithms could be applied.
5. Despite these variations, the data mining tasks in each complex data object are still being categorized into four main categories mentioned above.

Reflection

- What have you learned this week?

(Every student shall attempt to answer this question and you may be asked to provide your answer in the class.)